

Sentinels VR Hostage-Rescue Trainer · Module Evaluation

Hostage Emotional State Change Due to Gunfire Sound

A research-grounded reference for mapping sound-pressure level (dB), distance-to-source, and sound type onto the hostage/trainee emotional-state model (Calm → Fear → Freeze → Panic) used in `HostageController.cs`.

MODULE EVALUATION · RESEARCH REVIEW

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1. Purpose & Scope

This document consolidates published research on how sound — specifically gunfire — changes an average person's emotional state, and translates it into concrete thresholds usable for the hostage emotional-state finite state machine (`Calm` → `Fearful` → `Panic` / `Freeze`) in the VR hostage-rescue trainer. It answers three questions:

1. What emotional states can sound plausibly drive, and by what mechanism?
2. How loud (dB) is gunfire at a given distance from the hostage?
3. At what dB / distance does an average person's emotional state actually shift?

2. Emotional State Framework

2.1 Two modelling approaches

Model	Description	Best used for
Dimensional (Valence – Arousal – Dominance)	Continuous scale: pleasant↔unpleasant, calm↔excited, in-control↔overwhelmed. Operationalised in the IADS-2 database of 167 rated real-world sounds.	A continuous distress-index curve (matches the existing dashboard distress index).
Categorical (discrete states)	Ekman's "basic six" (fear, anger, disgust, sadness, surprise, happiness) collapsed, for combat purposes, to Calm → Alert → Fear → Panic → Freeze .	The existing <code>HostageState</code> enum.

2.2 The defense-response sequence (Bracha, 2004)

The most directly applicable model is Bracha's revision of "fight-or-flight," which orders the acute stress response as **Freeze** → **Flight** → **Fight** → **Fright** → **Faint**. The critical, counter-intuitive finding: **Freeze is the default first-stage reaction** to a sudden threat sound — not a deeper escalation past Fear. It is reflexive: reduced motion, dropped heart rate (bradycardia), raised muscle tone, and narrowed attention while the brain locates the threat. Flight, fight, or full panic only follow once the threat is appraised (escape route available? cornered? confirmed hostile?).

Design implication: Freeze should not require passing through Fear first. A sudden, close, unambiguous gunshot can plausibly jump `Calm` → `Freeze` directly, with Fear/Panic as the *conscious appraisal* that follows a beat later.

Bracha, H.S. (2004). *Freeze, Flight, Fight, Fright, Faint: Adaptationist Perspectives on the Acute Stress Response Spectrum*. CNS Spectrums. <https://www.cambridge.org/core/journals/cns-spectrums/article/freeze-flight-fight-fright-faint-adaptationist-perspectives-on-the-acute-stress-response-spectrum/6F80B347B1414CE066821D1620D6F8B1>

3. Sound-Pressure Fundamentals

3.1 Gunfire SPL is fixed at the source

A given firearm produces a consistent sound-pressure level (SPL) at a fixed reference distance every time it fires. Measured values from the literature:

Source	Measured peak SPL	Measurement point
Small arms (general)	140–180 dB	Varies by caliber/weapon
7.62 mm rifle	155–158 dB	At firer's ear
G3 rifle	161 dB peak	At firer's ear
M-72 LAW (rocket)	179 dB (firer) / 161 dB (8 m to the side/rear)	Firer's position vs. 8 m offset
120 mm mortar	185.5 dB peak	At firer's head
.40 pistol / .38 revolver (range)	111.9–116.8 dB(C)	At shooting stand

G.R. Price, U.S. Army Research Laboratory. *Weapon Noise Exposure of the Human Ear Analyzed with the AHAH Model*. https://arl.de.vcom.army.mil/wp-content/uploads/sites/3/2022/09/Weapon_noise_AHAAH.pdf | Noise of military weapons, JASA. <https://pubs.aip.org/asa/jasa/article/146/5/3832/994808> | Acoustic/psychoacoustic analysis of police firearms noise, PMC. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9450760/>

3.2 Sound loses 6 dB every time distance doubles

Standard acoustic inverse-square law (free field, no reflective surfaces):

$$\text{SPL}(d) = \text{SPL}(1\text{m}) - 20 \times \log_{10}(d) \quad [d \text{ in metres}]$$

This is confirmed by the real Army measurement above: the M-72 LAW dropped from 179 dB at the firer's position to 161 dB at 8 m — an 18 dB loss, consistent with the inverse-square prediction.

Sound Propagation – the Inverse Square Law. https://www.engineeringtoolbox.com/inverse-square-law-d_890.html

4. dB-by-Distance Table — Typical Rifle (160 dB at 1 m)

Using 160 dB at 1 m as a representative rifle/pistol muzzle SPL (mid-range of the 140–180 dB literature values above), applying the inverse-square law:

Distance from hostage to gunshot	Estimated SPL	Likely emotional state	Basis / citation
1 m (point-blank)	~160 dB	Panic / extreme Freeze–Fright	Injury-threshold range; Bracha's extreme "fright" tier
2 m	~154 dB	Panic / Fear	Well above occupational limit of 137 dB(C)
4 m	~148 dB	Fear , strong startle	Fear-potentiated startle range

8 m	~142 dB	Fear , likely reflexive Freeze first	Matches real Army-measured value (161 dB at 8 m for a rocket)
16 m	~136 dB	Freeze → Fear	Above the ~116 dB standard startle-stimulus level
32 m	~130 dB	Freeze (reflexive), fading toward Fear	Above the 93 dB acoustic-startle threshold
64 m	~124 dB	Alert → Freeze	Still above startle threshold, response less severe
128 m	~118 dB	Alert , borderline startle	At the edge of the 116 dB standard startle stimulus
250 m+	<100 dB	Calm , possible non-recognition	Distant/muffled shots are frequently misidentified as fireworks/construction

Acoustic startle chamber methodology. <https://conductscience.com/lab/acoustic-startle-chamber> | Fear-potentiated startle review, PMC. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6162305/> | Firearms occupational SPL limit, JASA. <https://pubs.aip.org/asa/jasa/article/151/3/1769/2838222> | Civilian gunfire misidentification, Iowa State ABM thesis. <https://www.imse.iastate.edu/files/2014/03/StewartAlex-thesis.pdf>

5. General dB → Emotional State Reference (Any Sound)

SPL (dB)	Example sound	State	Why / citation
< 40 dB	Quiet room, ambient	Calm	Below arousal/annoyance threshold – WHO night-noise guideline (<40 dB). https://www.who.int/europe/news-room/fact-sheets/item/noise
40–70 dB	Conversation, distant/muffled gunshot	Calm	Below annoyance threshold; distant gunshots often misidentified rather than feared. WHO guideline: https://cdn.who.int/media/docs/default-source/who-compendium-on-health-and-environment/who_compendium_noise_01042022.pdf
70–93 dB	Shouting, nearby alarm	Calm → rising Alert	WHO "highly annoyed" annoyance band, pre-startle. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6313593/
93–116 dB	Sudden nearby burst, standard lab startle stimulus	Freeze , then Fear	93 dB = acoustic startle threshold; 116 dB = standard startle-chamber stimulus. Freeze precedes conscious Fear (Bracha, 2004).
116–137 dB	Close alarm, near-range gunfire, occupational SPL ceiling	Fear , branching to Panic if inescapable	Fear-potentiated startle strongest here; Panic vs. sustained Freeze depends on threat appraisal, not loudness alone.
130–175 dB peak	Actual gunfire at engagement range	Fear / Panic , does not reliably habituate	Repeated exposure can fail to habituate, especially in trauma-sensitised listeners. https://pmc.ncbi.nlm.nih.gov/articles/PMC6387566/
> 160 dB	Close-range blast/muzzle	Panic / Fright–Faint (extreme)	Crosses pain/injury threshold, compounding fear with physical pain. https://pmc.ncbi.nlm.nih.gov/articles/PMC9450760/

6. Reaction by Sound Type (Not Just Loudness)

Sound	What an average person actually does	State	Citation
Distant/muffled single gunshot	Often not recognised as gunfire – misattributed to fireworks, a balloon popping, construction, a car backfiring. Real-incident recognition lag averages ~5 minutes.	Calm	Iowa State active-shooter ABM thesis; Active Response Training
Close, unambiguous gunshot	Reflexive, involuntary whole-body startle (eye-blink, muscle flinch) within ~40 ms, before conscious appraisal.	Startle → Fear	Acoustic startle reflex methodology; fear-potentiated startle review
Any sudden intense threat sound (first response)	Freeze is the default first reaction — not flight or fight. Motion stops, heart rate drops, attention narrows to locate the threat.	Freeze	Bracha (2004)
Threat confirmed, inescapable / cornered	Escalates to flight (exit available), fight (cornered), frantic "fright," or in extreme inescapable cases, fainting.	Panic	Bracha (2004)
Human scream nearby	Screams carry "roughness" (fast loudness modulation, 30–150 Hz) that hits the amygdala directly — judged more frightening than any other sound at equal loudness, including gunfire.	Sharp Fear spike	Arnal et al. (2015), <i>Current Biology</i> . https://www.cell.com/fulltext/S0960-9822(15)00737-X
Siren/alarm (rising pitch, fast pattern)	Engineered to maximise perceived urgency → primes flight readiness on first exposure. Repeated, non-escalating exposure causes " alarm tune-out " (habituation).	Alert/Panic → decays	Psychoacoustics of fire alarms, PMC7827080; alarm sounds & children's physiological response
Repeated gunfire on the same person	Most people habituate (startle drops each repeat). Trauma-sensitised individuals show the opposite: failure to habituate — fear stays constant or escalates.	Fear/Panic sustained	Blast-exposed startle-gating study, PMC6387566

Evidence-quality caveat: low-frequency/infrasound (<20 Hz) "dread" effects are frequently cited in sound-design contexts but remain scientifically contested — the foundational 19 Hz "haunted room" hypothesis (Tandy & Lawrence) has been disputed by later replications as confounded/suggestibility-driven. A 2023 controlled study found a measurable cortisol/stress increase from infrasound, but this is preliminary, not settled. Treat frequency-based dread effects as suggestive, not load-bearing, for module evaluation claims.

7. Application to HostageController.cs

- **Freeze is mis-ordered in most simple FSMs.** Freeze should be reachable directly from Calm on a sudden, close gunshot — not only as an escalation from Fearful at close range.
- **Recognition isn't instant.** A first, distant gunshot plausibly leaves the hostage at Calm rather than jumping straight to Fearful — real people frequently fail to identify a first distant shot as gunfire.
- **Screams outrank gunshots.** If a nearby NPC screams (e.g. the existing "get back!" hostage-scare mechanic), that is empirically a *sharper* fear trigger than a same-loudness gunshot — weight it higher than gunshot distance alone.
- **Habituation vs. sensitisation is a deliberate design choice.** Whether repeated gunfire in one session should decay the hostage's fear (normal habituation) or escalate it (trauma-sensitised / failure-to-habituate) is a defensible choice either way, depending on which population is being modelled.
- **Suggested distance bands** (from Section 4), to replace ad-hoc `freezeRange` / `panicRange` constants:

Range	Distance	Suggested state
Close	≤ 10 m	Panic / Fear (always)
Medium	10–60 m	Freeze first, then Fear
Far	60–150 m	Alert, mild Fear
Very far	> 150 m	Often stays Calm (non-recognition)

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